2D Arrays

1D array:

```
int[] a = {10, 20, 30};
```

2D array:

```
int[][] b = {
        {10, 20, 30},
        {40, 50, 60}    row 1
};
```

how do we access 50?

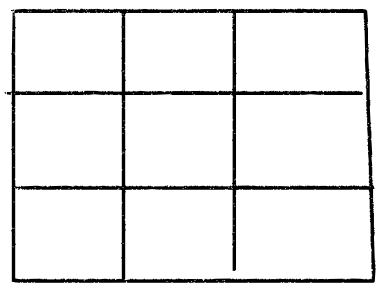
```
b[1][1]
```

2D Arrays

```
int[][] b = {
        {10, 20, 30},
        {40, 50, 60}    row 1
};
```

how do we change 50 → 80?

```
b[1][1] = 80;
```



Traverse a 2D Array

```
int[][] b = {
        {10, 20, 30},
        {40, 50, 60}
};
```

```
for(int raw = 0; raw < b.length; raw++
    for(int col = 0; col < b[raw].length; col++)
        println( b[raw][col] )
```

nested loop

Find the Average

```
int[][] b = {
        {10, 20, 30},
        {40, 50, 60}
    };

int sum = 0;
int totalElements = 0;

for(int raw = 0; raw < b.length; raw++
    for(int col = 0; col < b[raw].length; col++)
        sum += b[raw][col];
        totalElements += 1;
int avg = sum / totalElements;
```

Find the Max

```
int[][] b = {
        {10, 20, 30},
        {40, 50, 60}
    };
    int max = 0;
for(int raw = 0; raw < b.length; raw++)
    for(int col = 0; col < b[raw].length; col++)
        if( b[raw][col] > max) {
            max = b[raw][col];
        }
```

Count #s ≥ 65

```
int[][] b = {
        {10, 20, 30},
        {40, 50, 60}
    };

int count = 0;
for(int raw = 0; raw < b.length; raw++)
    for(int col = 0; col < b[raw].length; col++)
        if b[raw][col] ≥ 65 {
            count++;
        }
```